Ashish Kundu

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.32
Gaur Public School, Kaithal	Class XII (Board of School Education, Haryana)	2023	88.00%
Gaur Public School, Kaithal	Class X (Board of School Education, Haryana)	2021	100.00%

SKILLS

Tools: Solid Edge,**Visual Studio Code**, **Git, GitHub**, Canva, **Jupyter Notebook**, Aspen Plus

Languages: Proefficient in **C/C++**, familier with **Python**

Computer Science Fundamentals: **Data Structures and Algorithms(DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems (OS), and Computer Networks (CN)**

Soft Skills: Problem Solving Mindset, Awareness, Adaptability

PROJECTS

Smart Gym Equipment Manager | Python, Streamlit, SQLite, pytest (Link)

- Built a real-time equipment **queue system** using **round-robin scheduling** (quantum = 1 set) across 12+ equipment types, enabling members to rotate between sets instead of waiting out a full workout.
- Implemented the scheduling engine in pure **Python using deque-based queues** for **O(1) enqueue/dequeue operations** and a **hash-map index for O(1)** duplicate-queue checks, enforcing a strict one-equipment-per-user constraint.
- Added **SQLite**-backed user authentication with salted password hashing, along with date-scoped workout plans that persist reliably across application restarts.
- Designed the frontend using **Streamlit** for an intuitive, responsive interface, and validated core logic with pytest to ensure reliability under concurrent usage scenarios.

Resource Allocation Engine -- Conflict-Aware Scheduler (Link)

- Designed and implemented a **resource-allocation engine** that resolves overlapping requests for shared resources (GPUs, labs, rooms) using interval-based conflict detection and priority/deadline-aware scheduling, replacing naive reject-on-conflict logic with an optimal allocation strategy.
- Implemented and benchmarked **greedy vs. dynamic-programming**-based weighted interval scheduling (O(n log n) using binary search) to maximize total priority satisfied across competing requests, and proved greedy is provably suboptimal through a documented counterexample.
- Built a **REST API** using **Node.js/Express** with endpoints for resource registration, request intake, availability checks, and automated allocation, backed by **transactional locking** to prevent race conditions during concurrent bookings.
- Tested the **API** extensively for edge cases involving overlapping time windows and simultaneous requests to ensure allocation correctness and system robustness.

Splitwise like Expense Tracker :

- Built a **group expense-sharing application** using Python and Streamlit to record expenses, split costs among members, and track individual balances.
- Implemented **automated balance calculation and debt settlement**, determining who owes whom and the amount required to settle group expenses.
- Developed a **transaction minimization algorithm** to reduce the number of payments required for settling outstanding balances.

RELEVANT COURSES

Mathematics & Computing

- Calculus I & II, Differential Equations, Elementary Linear Algebra
- Numerical Methods- Solved ODEs and PDEs using Python
- Applied Mathematics in Chemical Engineering- Including various topics like **Statistics, Probability Machine Learning** using **Python**

Computer Science& Artificial Intelligence: Introduction to Programming[ID1063], **Artificial Intelligence**[ID1050]

Liberal Arts: Communication Skills, Ethics and Values

SCHOLASTIC ACHIEVEMENTS

Got **AIR 50** on National Level test of an Institution Allen in Class 8th.

Solved **400+ Questions** among all coding platforms like **LEETCODE, CODE360, GEEKSFORGEEKS**, and many more

POSITIONS OF RESPONSIBILITY

Core Member Cepheid: The **Astronomy** and **Astrophysics** Club of IIT Hyderabad: Contributed to organizing stargazing sessions, technical workshops, and outreach events to promote interest in astronomy among students as an active leader.

EXTRACURRICULAR

National Service Scheme (NSS) Volunteer: Actively participated in community service initiatives including cleanliness drives, awareness campaigns, and social outreach programs, contributing to civic engagement efforts.

Competitive Kabaddi Player: Won multiple District Level Kabaddi tournaments as Leading Captain and led the team to victory in the Chemical Engineering Department finals at the college level.

Regular Volleyball player, participating in inter-department tournaments.